

Data structures and algorithms

Data Structures and Algorithms (DSA) are fundamental concepts in Computer Science that teach you how to think and solve complex problems systematically. Using the right data structure and algorithm makes your program run faster, especially when working with large amounts of data.

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Graph Algorithms

Graph Traversal: Techniques for visiting all vertices in a graph.

Depth-First Search (DFS): Explores as far as possible along each branch before backtracking.

Breadth-First Search (BFS): Explores neighbor vertices before moving to the next level of vertices.

Minimum Spanning Trees: Minimum Spanning Trees are the smallest trees that connect all nodes in a graph without forming cycles, achieved by adding the shortest edges possible.

Kruskal's Algorithm: Finds a minimum spanning tree for a

connected weighted graph. It adds the smallest weight edge that does not form a cycle.

Prim's Algorithm: Also finds a minimum spanning tree for a connected weighted graph. It adds the smallest weight edge that connects two trees.

Topological Sorting: Topological sorting is a linear ordering of the vertices in a directed acyclic graph such that for every directed edge from vertex u to vertex v , u comes before v in the ordering.

Kahn's Algorithm: Finds a topological ordering of a directed acyclic graph.

DFS-based Algorithm: Uses Depth-First Search to perform topological sorting in a directed acyclic graph.

Shortest Path: A shortest path in a graph is the path between two vertices that has the minimum sum of weights along its edges compared to all other paths between the same two vertices.

Dijkstra's Algorithm: Greedy algorithm to find the shortest path between all nodes in $O(\text{time})$.

Floyd-Warshall Algorithm: Finds the shortest path between all pairs of nodes in $O(\text{time})$.

Bellman Ford Algorithm: Finds the shortest path from a single source in $O(\text{time})$.

Johnson Algorithm: Finds the shortest paths between all pairs of vertices in $O(\text{time})$.

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Asymptotic Notation

Big-O: Describes the worst-case scenario, providing an upper time bound of an algorithm.

Omega: Describes the best-case scenario, offering a lower time bound of an algorithm.

Theta: Represents the average complexity of an algorithm.

Mathematical Algorithms

GCD and LCM: Find the greatest common divisor and least common multiple of two numbers.

Catalan Numbers: Count the number of valid expressions with a given number of pairs of parentheses.

Euler Totient Function: Count the number of positive integers less than a given number that are relatively prime to it.

nCr Computations: Calculate the binomial coefficient, which represents the number of ways to choose r elements from a set of n elements.

Prime Numbers and Primality Tests: Determine whether a given number is prime and find prime numbers efficiently.

Geometric Algorithms

Closest Pair of Points: Finds the two points in a set that are closest to each other.

Line Intersection: Determines whether two lines intersect and, if so, finds the point of intersection.

Greedy Algorithms

Fractional Knapsack: Find the maximum total value of items that can be placed in the knapsack, allowing fractional inclusion of items.

Dijkstra's Algorithm: Finds the shortest path from a source vertex to all other vertices in a weighted graph.

Huffman Coding: Creates an optimal prefix code for a set of symbols, minimizing the total encoding length.

Randomized Algorithms

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Skip List: A randomized data structure that provides fast search and insertion operations.

Dynamic Programming

Longest Common Subsequence: Finding the longest

subsequence common to two sequences.

Longest Increasing Subsequence: Finding the longest subsequence of a given sequence in which the elements are sorted in increasing order.

0/1 Knapsack Problem: Determining the maximum value that can be obtained by selecting items with given weights and values, while not exceeding a specified weight limit.

Rod Cutting Problem: Maximizing the profit by cutting a rod of given length into pieces and selling them according to given prices.

Coin Change Problem: Determining the number of ways to make change for a given amount using a given set of coin denominations.

- **Edit Distance:** Finding the minimum number of operations required to convert one string into another.
- **Minimum Cost Path:** Finding the minimum cost path from the top-left corner to the bottom-right corner of a given grid.

Remember, Data Structures and Algorithms are essential skills for any computer science student or professional. Mastering these concepts will help you write efficient and optimized code, making you a better programmer.

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